8 **(a)** Describe how the alpha scattering experiment provides evidence for

(i) the small size of the nucleus,

……

……[1]

(ii) a charged nucleus.

……

……[1]

(b) In 1914, James Chadwick showed that the energies of the beta particles emitted for a radioactive source had a distribution of energies rather than with a distinct single value of energy.

Figure 8.1 shows the energy spectrum for beta particles emitted during the decay of Bismuth-210 ($^{210}_{83}Bi$). The intensity (vertical axis) indicates the number of beta particles emitted with each particular kinetic energy (horizontal axis).

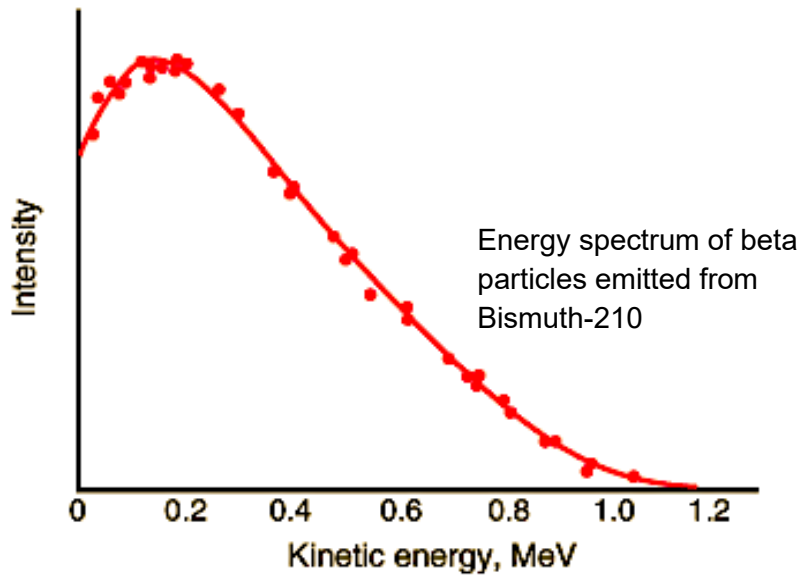

Fig. 8.1

(i) **1.** Determine, from Fig.8.1, *Q*, the maximum possible energy of the beta particle emitted.

Q =…… MeV [1]

2. Hence calculate the maximum speed of the beta particle.

Maximum speed =…… m s^{-1}

3. Comment on the value you obtained in **b(i)2**.

……

……[2]

8 **(b)** **(ii)** The radioactive isotope of Bismuth, $^{210}_{83}Bi$, decays into Polonium (chemical symbol: Po) with the emission of a beta particle.

Determine the mass of the resultant Polonium nucleus, in terms of *u*, and express your answer to 3 decimal places. (mass of a $^{210}_{83}Bi$ nucleus is 209.939 *u*; mass of proton m_p is 1.00729 *u*; mass of neutron m_n is 1.00867 *u*).

mass=…… *u* [3]

(iii) From Fig. 8.1, identify the most probable energy for the beta particle.

most probable energy value = …… MeV [1]

(iv) It is noted that the stable isotopes of heavy elements have an optimal neutron to proton ratio. Unstable isotopes will undergo transmutation into another element through radioactive decay such that product achieve the optimal ratio.

Suggest, with a reason, whether Bismuth-210 has an excess of neutrons or protons, as compared to the optimal ratio.

……

……

……

……[2]

(v) The continuous spectrum of kinetic energy values of the beta particle presented a problem to physicists up to 1930s. If a stationary nucleus decayed into a beta particle and a stable daughter nucleus only, it should lead to a distinct single value of energy.

Explain, using conservation of linear momentum and energy, how the continuous spectrum of beta particle energies gave rise to this problem.

……

……

……

…… [2]

8 **(b)** **(vi)** Suggest what was proposed by physicists to resolve the problem in **(v)**.

……

……[1]

(c) In an experiment, a detector is held a fixed distance from a sample of a radioactive material and the data provided is used to plot a graph of count-rate against time, as shown in Fig 8.2.

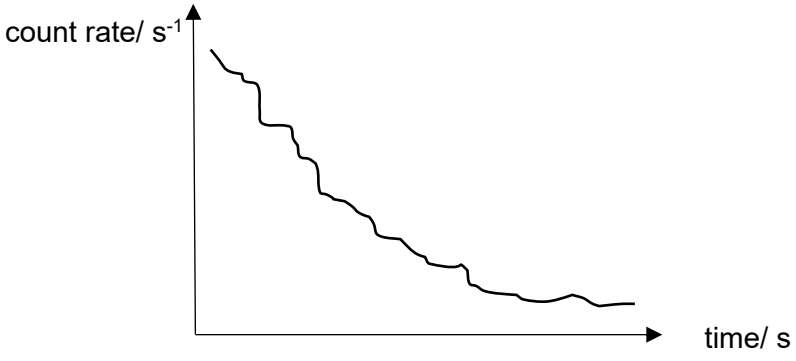

Fig. 8.2

(i) Explain why the data points on the graph do not lie on a smooth curve.

……

……[1]

(ii) Suggest two reasons why the count-rate recorded is not equal to the activity of the radioactive sample.

1. ……

……

2. ……

……[2]

(iii) The count rate from a counter near a radioactive source is $7.6 \times 10^{8}\ s^{-1}$. The decay constant of the source is $4.6 \times 10^{-3}\ s^{-1}$.

1. Calculate the time taken for the count rate to fall to $8.3 \times 10^{3}\ s^{-1}$.

time = …… s [1]

8 (c) (iii) 2. When the detector is at a distance *y* from the radioactive source, the count rate is 234 counts per minute.

Calculate the count rate when the detector is at a distance 3*y* from the source.

average count rate = ……. counts per minute [2]